

DTRA MASTERCLASS

PART 1: THE CRISIS

The Alert Fatigue Epidemic

The Hard Numbers (2025 Data)

SOC Analyst Burnout Crisis

Statistic	Value	Source (2025)
SOC analysts reporting burnout	71%	Dark Reading Survey 2025
Organizations reporting analyst burnout	73%	Cybersecurity Insiders 2025
Security professionals experiencing exhaustion	76%	Abnormal AI Report 2025
Analysts considering leaving due to burnout	65%	SecurityMetrics 2025
Alert fatigue cited as top challenge	76%	Cybersecurity Insiders 2025

Alert Fatigue Statistics

Statistic	Value	Source (2025)
Organizations where alert volume increased	88%	Cybersecurity Insiders 2025
False positives identified as main challenge	73%	Stamus Networks 2025
Security alerts being false positives	>50%	Torq Research 2025
Alerts missed due to fatigue	30%	Databahn AI 2025
Organizations processing 960+ alerts daily	Average	Hacker News Report 2025
Analysts muting alarms to cope	40%	RedCarbon AI 2025

Financial Impact

IBM Cost of Data Breach Report 2025:

Metric	Value
Global average breach cost	\$4.44 million
US average breach cost (all-time high)	\$10.22 million
Cost savings with AI/automation	\$1.9 million
Breach lifecycle reduction with AI	80 days faster
Healthcare sector (highest)	\$7.42 million
Malicious insider breaches	\$4.92 million

Key Insight: Organizations using AI extensively cut breach lifecycle by 80 days and saved nearly \$1.9 million average.

Why Detection Is Solved But Decision Is Not

The Paradox

Better Detection Technology → More Alerts Generated → Analyst Overload → Slower Response → Missed Threats

Detection Evolution

Era	Technology	Detection Accuracy	Problem Created
1990s	Signature IDS	~70%	Missed novel attacks
2000s	Anomaly Detection	~85%	High false positive rate
2010s	ML-based IDS	~95%	Alert volume explosion
2020s	Deep Learning IDS	~99%	Analyst burnout epidemic

The Unsolved Problem

"Detection technology has advanced dramatically. What hasn't advanced is human capacity to triage 100,000+ daily alerts."

2025 Reality: A single day's alerts could take 61+ days to fully investigate (Abnormal AI 2025).

What Industry Tried (And Why It's Not Enough)

Solution Type	What It Does	Why It Fails at Decision
IDS (Snort, Suricata)	Detects threats, generates alerts	Zero response capability
IPS	Blocks based on static rules	No learning, no adaptation
SIEM (Splunk, QRadar)	Aggregates and correlates logs	Still requires human analysis
SOAR	Automates playbooks	Playbooks are static rules
AI-Enhanced IDS	ML for better detection	Response still rule-based

Common Thread Across All Solutions:

- No reinforcement learning from outcomes
- No context-aware decision making
- No continuous improvement after deployment

PART 2: OUR THESIS

The Scientific Foundation

DTRA's Core Innovation

The Key Distinction

Aspect	IDS/IPS Approach	DTRA Approach

Primary Aspect / Function	Threat Detection IDS/IPS Approach	Autonomous Triage Decision DTRA Approach
Output	Alert	Decision (Block/Log/Ignore)
Intelligence	Pattern matching	Reinforcement Learning
Adaptation	Manual rule updates	Learns from every decision
Human Role	Required for every alert	Reviews only uncertain cases

Architecture Paradigm

TRADITIONAL:	Network → IDS → [100,000 ALERTS] → [HUMAN] → Decision
DTRA:	Network → Hybrid AI → Danger Score → [Q-AGENT] → Decision → [Dashboard oversight]

The Claim

"DTRA is not a better IDS. It is an autonomous Tier 1 SOC analyst that handles routine triage decisions, escalating only uncertain cases to humans."

The A²C Framework - Our Theoretical Foundation

From ACM Transactions on Internet Technology (2024-2025):

The A²C Framework defines three modes of human-AI SOC operation:

Mode	Definition	DTRA Implementation
Automation	AI handles routine tasks	Q-Learning agent processes 80% of alerts
Augmentation	AI provides context to humans	Dashboard shows danger scores + SHAP explanations
Collaboration	Human + AI on complex cases	Analyst reviews escalated alerts, can override

Academic Grounding

"By implementing and operationalising A²C, SOC's can significantly reduce alert fatigue while empowering analysts to efficiently and effectively respond to security incidents."

We are implementing a published, peer-reviewed framework - not speculative research.

Why Reinforcement Learning? (DARPA Validation)

DARPA Programs Supporting This Approach

1. DARPA CASTLE (Cyber Agents for Security Testing and Learning Environments)
 - Uses RL agents for autonomous cyber defense
 - Oak Ridge National Lab: "proper reward functions achieve machine-speed defense"
2. DARPA AI Cyber Challenge (AixCC)
 - Team Atlanta won **\$4 million**
 - Approach: Multi-agent RL for autonomous threat triage
 - Same core technique we employ

Why Q-Learning Specifically

Approach	Advantages	Disadvantages	Our Choice
Rule-based	Fast, interpretable	Static, doesn't learn	❌
Deep RL (DQN)	Handles complex states	Black box, needs massive data	Future

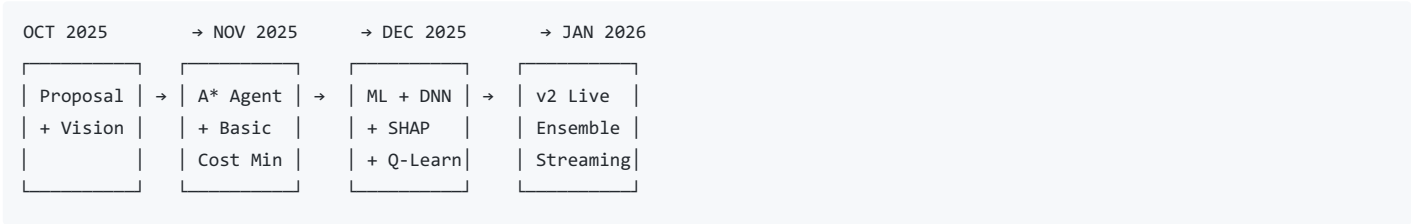
Approach	Advantages	Disadvantages	Our Choice
Tabular Q-Learning	Interpretable, fast, learns from outcomes	Limited state space	☒

Rationale: Q-Learning provides interpretability (can explain Q-table to stakeholders), fast convergence, and proven effectiveness for discrete action spaces.

PART 3: THE EVOLUTION

From Concept to v2

Project Evolution Timeline



Key Evolution Milestones

Stage	What We Built	Key Learning
Proposal	Vision: Hybrid AI + RL for alert fatigue	Problem statement validated
A Agent*	Cost-minimization decision logic	Static rules don't adapt
ML Baseline	94% accuracy, 788 false positives baseline	Need DNN + ensemble
DNN + RL	98% accuracy, 47% FP reduction, Q-Learning	A* replaced with RL for adaptability
v2	Modern dataset, live streaming, stacked ensemble	Production-ready architecture

Critical Pivot: A* → Q-Learning

A* (Static)	Q-Learning (Adaptive)
Calculates cost once	Updates Q-table with every decision
Same input = same output always	Learns which decisions worked
Cannot adapt to new patterns	Gets smarter continuously

Why the change: We realized static cost functions can't account for outcome feedback. Q-Learning learns optimal policy through experience.

Results Progression

Detection Accuracy

Stage	Model	Accuracy	Recall	False Positives
Baseline	Logistic Regression	94%	83%	788 / 30K

Advanced Stage	Deep Neural Network Model	98% Accuracy	96% Recall	412 / 30K False Positives
v1 Final	DNN + Random Forest	98.9%	96%	-47% reduction
v2	DNN + XGBoost (stacked)	96.05%	92.34%	Modern dataset

Q-Learning Learned Policy

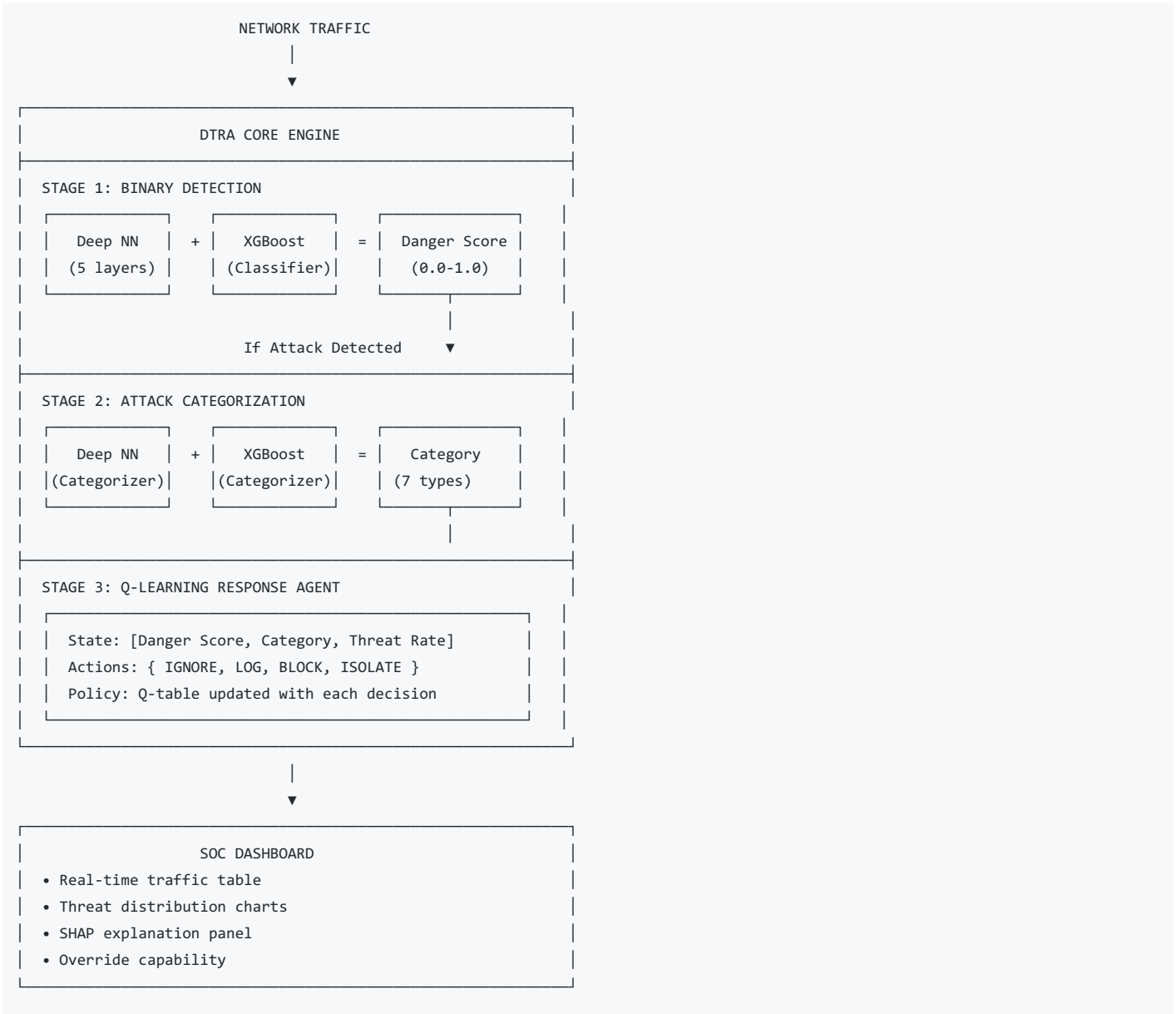
Danger Level	Learned Action	Rationale
0-30% (Low)	IGNORE	Avoid FP penalty, preserve analyst time
30-70% (Medium)	BLOCK	Risk of ignoring > disruption of blocking
70-90% (High)	ISOLATE	Maximum threat = maximum containment
90-100% (Critical)	BLOCK	Isolate too costly; Block is sufficient

Nuance discovered: Agent learned that even 90-100% threats should be BLOCKED not ISOLATED - the business disruption of isolation outweighs marginal security benefit. This is learned optimization A* could never discover.

PART 4: TECHNICAL ARCHITECTURE

Engineering Decisions

System Architecture



Why Hybrid Ensemble (DNN + XGBoost)?

2025 Research Validation

From Hybrid IDS Research (2025):

- Hybrid detection systems achieve **97.8% accuracy** with **2.1% false positive rate**
- Ensemble models using RF-RFE report **99%+ accuracy** on benchmark datasets
- Hybrid approaches show **65% improvement** in detection speed

Model Combination Logic

Model	Strength	Weakness
Deep Neural Network	Complex non-linear patterns	Overfits, black box
XGBoost	Robust on tabular data, handles imbalance	Less effective on complex patterns
DNN + XGBoost Ensemble	Best of both worlds	Acceptable compute overhead

Soft Voting Implementation

```
ensemble_probs = (dnn_probs + xgb_probs) / 2
final_prediction = argmax(ensemble_probs)
```

Q-Learning Implementation

The Q-Learning Formula

$$Q(s,a) \leftarrow Q(s,a) + \alpha [R + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$$

Symbol	Meaning	Our Value
s	Current state (danger level)	[Low, Med, High, Critical]
a	Action taken	{Ignore, Log, Block, Isolate}
R	Immediate reward	+10 correct, -50 missed attack, -20 false positive
α	Learning rate	0.1
γ	Discount factor	0.95

Reward Function Design

Scenario	Reward	Rationale
Correctly block real attack	+10	Security win
Correctly ignore false positive	+5	Efficiency win
Miss real attack	-50	Security failure (heavy penalty)
Block false positive	-20	Business disruption

Why Reward Asymmetry?

Missing an attack (-50) is penalized more than false positive (-20) because:

- Missed attacks can cause \$4.44M+ breach cost
- False positives cause frustration but no breach
- This makes the agent "security-first" but not paranoid

SHAP Explainability

Global Feature Importance

Rank	Feature	Contribution
1	Flow Duration	High
2	Bwd Packet Length Max	High
3	FIN Flag Count	Medium
4	Fwd IAT Standard Deviation	Medium

5 Rank	Destination Port Feature	Medium Contribution
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Local Explanation Example

For a predicted **MALWARE** alert (confidence: 0.87):

"Classified as malware because:"

- FIN Flag Count = 0 (unusual for normal traffic) → +0.23
- Bwd Packet Length Max = 12,847 bytes (abnormally large) → +0.19
- Flow Duration = 0.3 seconds (too short for normal session) → +0.15"

Why This Matters

Without SHAP	With SHAP
"Alert: Possible malware"	"Alert: Malware - Large inbound packets with no FIN handshake"
Analyst investigates blindly	Analyst knows exactly what to look for
Trust: Low	Trust: High

Class Weights vs. SMOTE Decision

The Imbalance Problem

Real network traffic distribution:

- ~80% benign traffic
- ~20% malicious
- Some attack types <1% of total

Two Approaches

SMOTE	Class Weights (Our Choice)
Creates synthetic minority samples	Penalizes missing rare attacks
Balances training data	Keeps real distribution
Risk: Fake data → inflated accuracy	Honest accuracy on real data
May fail in production	Production-ready

Our Implementation

```
class_weights = compute_class_weight('balanced', classes=unique(y), y=y)
# Example: Benign: 0.6, DDoS: 1.2, Malware: 8.5 (rare = higher penalty)
```

Principle: 92% accuracy on real data > 99% on synthetic data.

PART 5: VALIDATION & COMPARISON

Proving Our Work

Our Results vs. Industry Benchmarks

Detection Performance

System	Method	Accuracy	Recall	Source
Traditional IDS (Snort)	Signatures	~70%	~60%	Benchmarks
ML-based IDS	RF/SVM	~95%	~85%	Published papers
DL-based IDS	CNN/LSTM	~98%	~95%	2025 research
DTRA v2	DNN+XGB Ensemble	96.05%	92.34%	Our results
Hybrid Ensemble (2025 papers)	Various	97-99%	95%+	ResearchGate 2025

Industry Comparison

System	Type	Learning Capability	Decision Making
Snort	IDS	None	None (human decides)
Suricata	IDS/IPS	None	Rule-based
Darktrace	AI SOC	Unsupervised	Limited auto-response
Splunk SOAR	Automation	Workflow learning	Static playbooks
DTRA	RL Agent	Q-Learning from outcomes	Autonomous triage

The Differentiator

"Show me another commercial system with a Q-table that learns from its own blocking decisions."

The 80% Workload Reduction Math

Traditional SOC Workflow

Daily Alerts:	100,000
Time per alert:	2 minutes (triage + decision)
Total analyst time:	200,000 minutes = 3,333 hours/day
Analysts needed:	3,333 ÷ 8 = 417 analysts (8-hour shifts)
Reality:	SOCs have 5-15 analysts → BURNOUT

DTRA Workflow

Daily Alerts:	100,000
DTRA auto-handles (80%):	80,000 alerts = 0 human time
Escalated to analyst (20%):	20,000 alerts
Human time needed:	20,000 × 2 min = 667 hours/day
Analysts needed:	667 ÷ 8 = 83 analysts
WORKLOAD REDUCTION: (3,333 - 667) / 3,333 = 80%	

Financial Translation

Without DTRA: Understaffed SOC → missed threats → potential \$4.44M breach
With DTRA: Right-sized team + AI triage → faster response → breach prevented

PART 6: PATH FORWARD

Research-Driven Improvement

Our Current Plan

Phase 1: Literature Deep Dive (Current)

Reading ~20+ papers to understand:

- Gap analysis in autonomous SOC research
- State-of-art in RL for cybersecurity
- Ensemble architectures for IDS
- Real-world deployment challenges

Goal: Identify specific research gaps we can address.

Phase 2: Gap-Driven Improvement

Based on literature review, potential directions:

- Multi-agent RL (MARL) for distributed networks
- Adversarial robustness testing
- Transfer learning across environments
- Drift detection for evolving attacks

Goal: Direct our technical improvements toward validated research needs.

Phase 3: Validation & Publication

- Collect real-world performance metrics
- Submit paper to relevant venue (IEEE INMIC, IBCAST, or similar)

Goal: Peer-reviewed validation of our approach.

Phase 4: Iteration Based on Feedback

- Incorporate academic feedback
- Address identified limitations
- Build toward production-ready system

Current System Capabilities

What We Have Built

- ☑ Hybrid ensemble detection (DNN + XGBoost)
- ☑ Two-stage classification (binary + 7 categories)
- ☑ Q-Learning response agent with learned policy
- ☑ Live traffic streaming capability
- ☑ Real-time SOC dashboard
- ☑ SHAP explainability integration

What Remains to Explore

- Multi-agent architecture for scale
 - Adversarial attack resilience
 - Long-term drift detection
 - Integration with commercial SIEMs
 - Real-world pilot deployment
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APPENDIX: Reference Data

Key Statistics Card

2025 Crisis Numbers

Statistic	Value	Source
SOC burnout rate	71%	Dark Reading 2025
Alert fatigue as top challenge	76%	Cybersecurity Insiders 2025
False positives as main issue	73%	Stamus Networks 2025
Alerts missed due to fatigue	30%	Databahn AI 2025
Global breach cost	\$4.44M	IBM 2025
US breach cost (record)	\$10.22M	IBM 2025
Cost savings with AI	\$1.9M	IBM 2025
Breach lifecycle reduction with AI	80 days	IBM 2025

Our Results

Metric	Value
Dataset	CIC-IIoT 2025
Binary Detection Recall	92.34%
Category Accuracy (7 classes)	92.83%
False Positive Reduction vs Baseline	47%
Architecture	DNN + XGBoost (stacked ensemble)

Academic Citations

- "Alert Fatigue in SOCs: Research Challenges" - ACM Computing Surveys 2025
- "A²C Framework for Human-AI SOC Teaming" - ACM Trans. IT 2024-2025
- "Autonomous Cyber Defense Agents" - ACM CSET (DARPA-aligned) 2024
- "Hybrid Ensemble for Intrusion Detection" - ResearchGate 2025
- "MARL in Cybersecurity" - arXiv 2025

One-Slide Summary

DTRA IN ONE SLIDE	
PROBLEM:	71% of SOC analysts face burnout 73% cite false positives as main challenge 30% of alerts missed due to fatigue
GAP:	Detection is solved. Decision is not. Existing tools detect but don't decide. Humans can't scale to 100K+ daily alerts.
SOLUTION:	DTRA = Hybrid AI Detection + Q-Learning Decision Agent Autonomous Tier 1 triage → 80% workload reduction
VALIDATION:	Implements A²C Framework (ACM 2024-2025) Aligns with DARPA CASTLE program 92%+ accuracy on modern IoT dataset
EVOLUTION:	Proposal → A* → ML → DNN+SHAP+RL → v2 Live Ensemble Key pivot: A* (static) → Q-Learning (adaptive)
PATH:	Literature review → Gap identification → Improvement → Validation → Publication